FUEL INJECTOR [SKYACTIV-D 1.5]

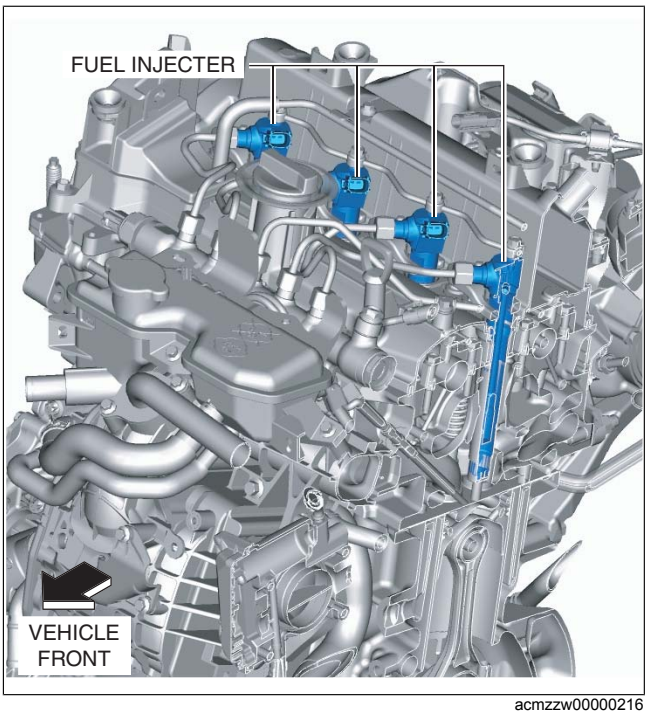
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Purpose, Function

- The fuel injectors inject fuel in the cylinder over a wide area, intake air and fuel are mixed efficiently, and the optimum air-fuel mix are formulated according to the driving conditions.

Construction

- The fuel injectors are located on the cylinder heads.
- Solenoid-type injectors have been adopted for the fuel injectors.



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Operation

Fuel injection mechanism

- Fuel is injected by opening/closing the control valve based on the ON/OFF signal from the PCM. The fuel injection time is determined by the energization time for the solenoid. The injection times can be controlled by intermittently energizing the solenoid.

  Before injection (while injection stopped)

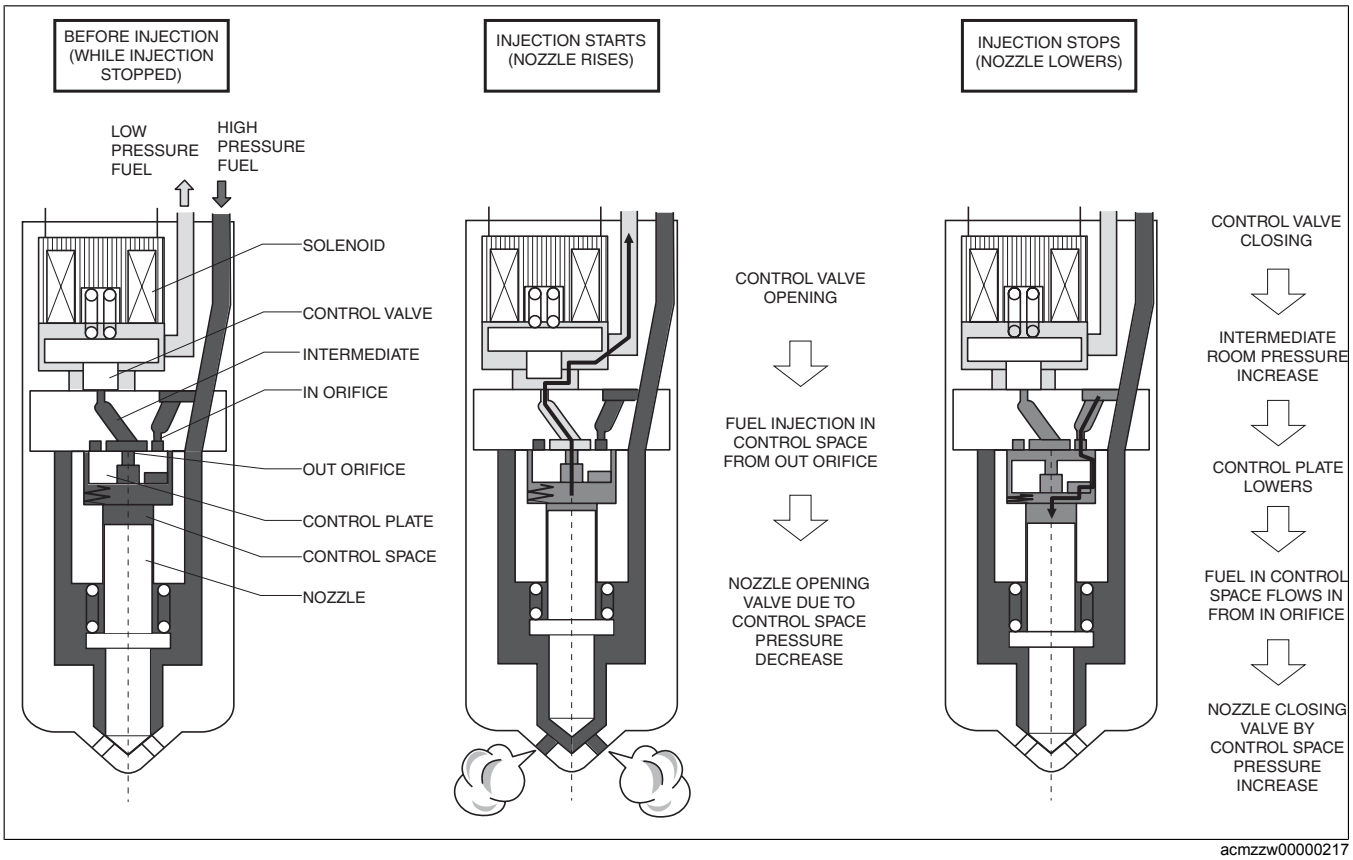
  — Because the control valve is pressed down when there is no energization to the solenoid, the passage sending fuel from the intermediate space is closed.Because the pressurized fuel sent from the injection pipe fills the control space, and pressure for pressing down the nozzle is applied, fuel injection is not performed.

  Injection starts (nozzle rises)

  — When energization to the solenoid starts, the control valve is pulled up and fuel in the control space passes through the OUT orifice and flows out.
  When the fuel flows out, the pressure of the control space decreases, the nozzle is pressed up, and fuel injection is started.

  Injection stops (nozzle lowers)

  — When energization to the solenoid stops, the control valve is pressed down, the pressure in the intermediate space increases, and the control plate is pressed down.
  The IN orifice is opened by the lowering of the control plate, and the high pressure fuel flows into the control space.
  The nozzle is closed instantaneously by increasing the pressure in the control space and the fuel injection is stopped.



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- For the fuel injection control, refer to the CONTROL SYSTEM. (See FUEL INJECTION AMOUNT CONTROL [SKYACTIV-D 1.5].) (See MULTIPLE FUEL INJECTION CONTROL [SKYACTIV-D 1.5].) (See FUEL INJECTION TIMING CONTROL [SKYACTIV-D 1.5].)

Fail-safe

DTC No.	Fail-safe function
P0201:00	• Limits the engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function. • PCM restricts engine-transaxle integration control. (ATX)
P0202:00	
P0203:00	
P0204:00	
P02EE:00	
P02EF:00	
P02F0:00	
P02F1:00	
P2146:00	
P2147:00	
P2148:00	
P2149:00	
P2150:00	
P2151:00	
P1378:00	• Limits the engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Fully open the intake shutter valve. • Inhibits engine-stop by operating the i-stop function. • PCM restricts engine-transaxle integration control. (ATX)
P1379:00	
P1675:00	
P1676:00	